Depth First Search (DFS)

```
#include <stdio.h>
#define MAX 50

int graph[MAX][MAX];
int visited[MAX];
int n;

void DFS(int vertex)
{
    int i;

    // Mark vertex as visited
    visited[vertex] = 1;

    // Display vertex
    printf("%d ", vertex);

    // Visit adjacent vertices
    for (i = 0; i < n; i++)
    {
        if (graph[vertex][i] == 1 && visited[i] == 0)
        {
            DFS(i);
        }
    }
}

int main()
{
    int i, j;
    int start;

    // Read number of vertices
    printf("Enter number of vertices: ");
    scanf("%d", &n);

    // Read adjacency matrix
    printf("Enter the adjacency matrix:\n");
    for (i = 0; i < n; i++)
    {
        for (j = 0; j < n; j++)
        {
```

```
            scanf("%d", &graph[i][j]);
        }
    }

    // Initialize visited array
    for (i = 0; i < n; i++)
    {
        visited[i] = 0;
    }

    // Read starting vertex
    printf("Enter starting vertex: ");
    scanf("%d", &start);

    printf("DFS Traversal: ");
    DFS(start);

    return 0;
}
```

```
Output:
Enter number of vertices: 5
Enter the adjacency matrix:
0 1 1 1 0
1 0 1 0 0
1 1 0 0 1
1 0 0 0 0
0 0 1 0 0
Enter starting vertex: 0
DFS Traversal: 0 1 2 4 3
```